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Wi-Fi HaLow module user guide

Table of Contents

1. INTRODUCTION 3

2. PROCEDURE 4

 2.1 Pre-Requisites 4

 2.2 Setup 5

 2.3 Wi-Fi HaLow Initial Configurations..... 11

 2.4 Access Point and Station setup 14

 2.5 Server and Client setup 16

 2.6 Latency Test 17

3. APPENDIX 18

 3.1 JP Bands for Wi-Fi HaLow..... 18

 3.2 Program the image on Sony Spresense 19

 3.3 Wi-Fi HaLow Card Console 21

Revision History

Rev	Date	Description of Change
1	06 July 2023	First Release

1. INTRODUCTION

The Wi-Fi HaLow board is powered by the NRC9272-industry's first system on a chip (SoC) that is compliant with the IEEE 802.11ah Wi-Fi HaLow standard. Wi-Fi HaLow is optimized Wi-Fi solution for internet of things (IoT) devices. Operating in the Sub 1GHz license-exempt band, it offers a much greater range and material penetration when compared to 2.4GHz and 5GHz technologies. It can support low data rate sensors up to high data rate surveillance camera applications. The protocol's power-saving mechanisms greatly reduce power consumption and increase battery life.

This module is compatible with **Sony's SPRESENSE** main board which is an IoT board computer that combines low power consumption with a GPS reception function and a high-resolution audio codec. This module has the interfaces of microphone input, headphone output, and microSD card slot to use with the SPRESENSE

2. PROCEDURE

This document details the information on how to use this Wi-Fi HaLow board along with Sony's SPRESENSE main board and how to configure them as either an Access Point or the Station and transfer the data between them.

2.1 Pre-Requisites

1. Please prepare the SPRESENSE board with required bootloader and nuttx spk files

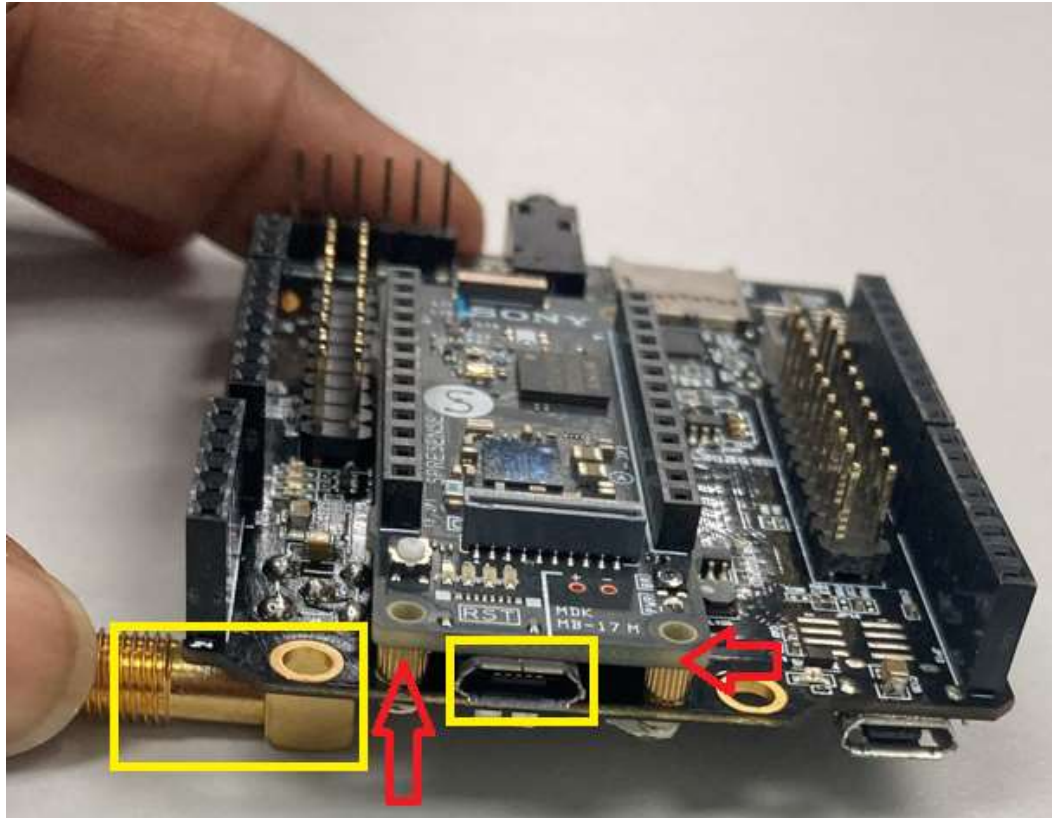
Please follow the below link to install the loader and nuttx.spk in SPRESENSE

https://developer.sony.com/spresense/development-guides/sdk_set_up_ide_en.html#_setup_for_windows_10_for_msys2

Else please refer the Appendix Section 3.2 in this document to program the bootloader/ nuttx.spk

2.2 Setup

2. Please insert the Sony SPRESENSE board as shown below
 - a) Please align the SPRESENSE's USB port on RF connector side of the Wi-Fi HaLow board (highlighted in Yellow)
 - b) Please align the 4 mounting holes on SPRESENSE board on the respective standoffs in the Wi-Fi HaLow card.
 - c) Then firmly press the main board down thus SPRESENSE mates with Wi-Fi HaLow card.

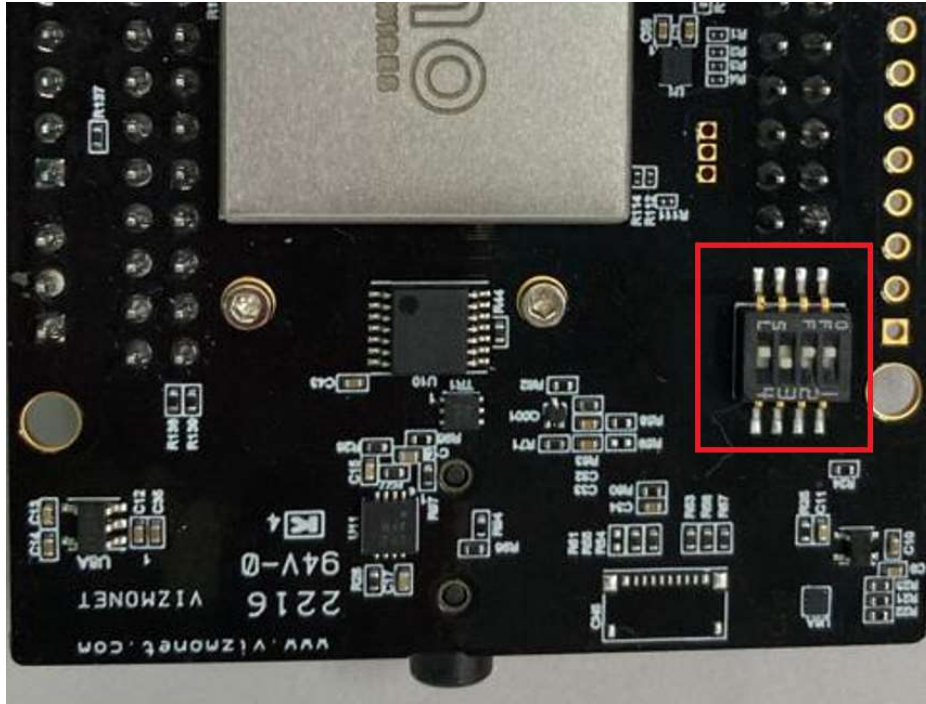


3. Please ensure the Wi-Fi HaLow board is in the ATCMD mode by keeping the switch position to below

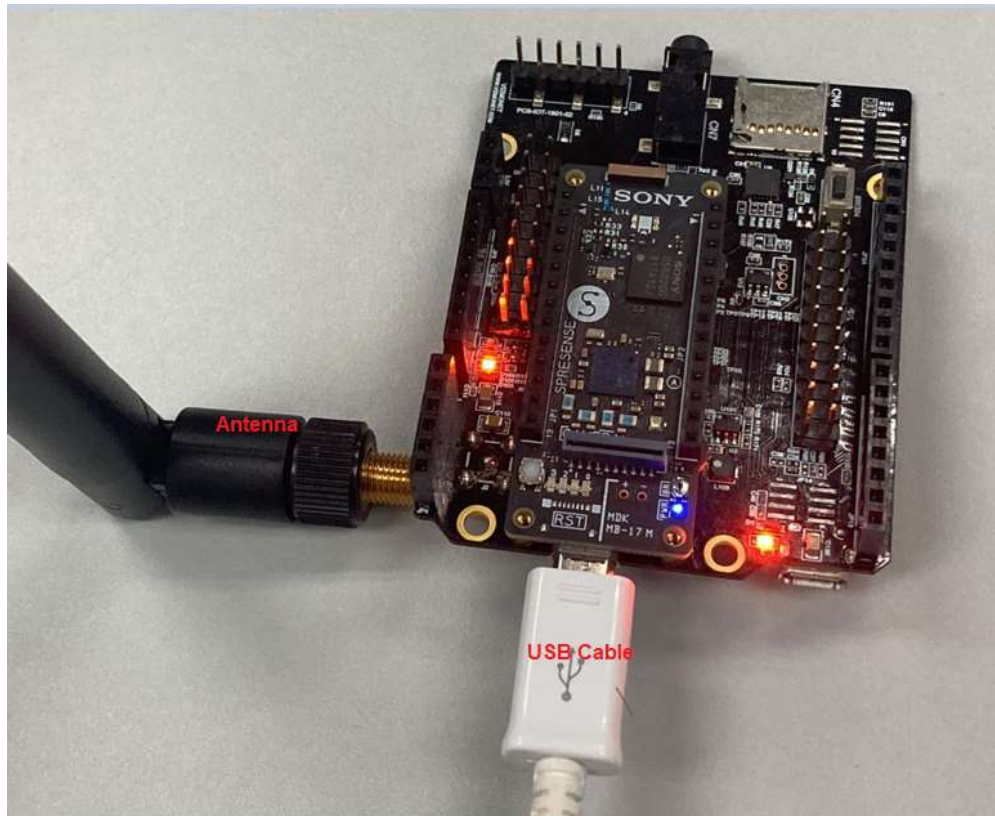
Pin1=OFF

Pin2=OFF

Pin3=ON

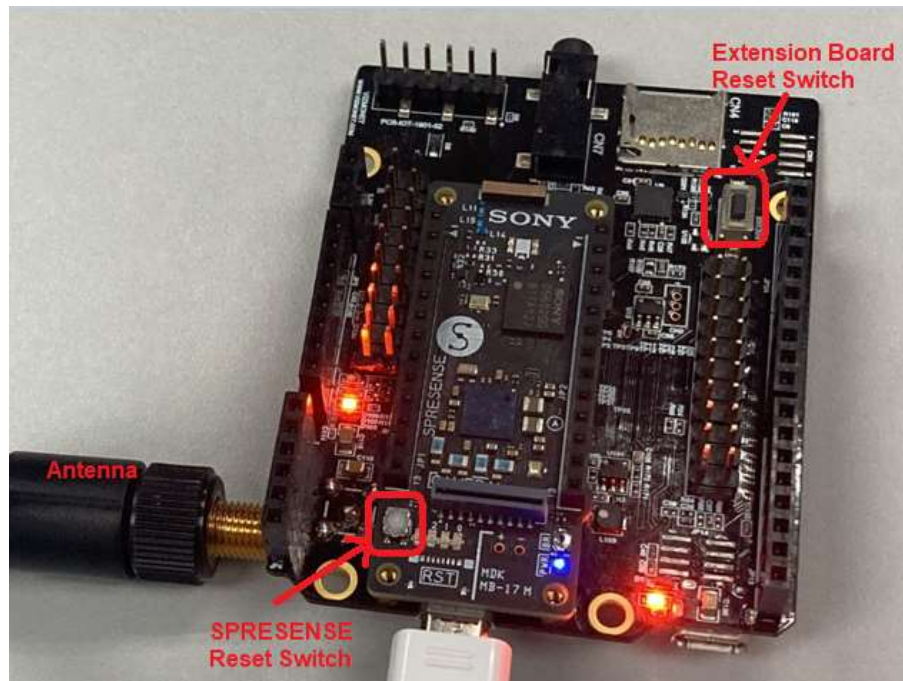


4. Please connect the Antenna with RF output connector of Wi-Fi HaLow board as shown in below figure
5. Please Power ON the module by connecting the USB cable with the Sony Spresense Main board it as shown in below figure (This USB cable will act as power cable as well as console for the this module)



This USB console cable in this figure will be used to communicate with Sony SPRESENSE board which will be issuing the AT commands to the Wi-Fi HaLow card through the SPI lines.

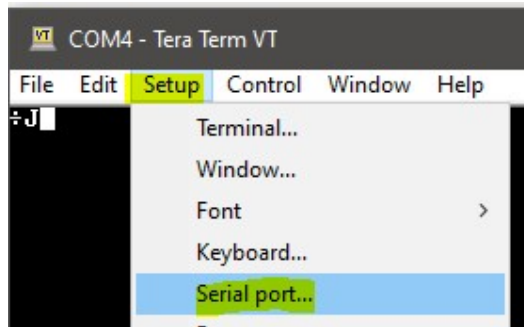
6. There are two reset switches, one for Wi-Fi HaLow board and one for Sony SpreSense as shown in figure. If required, they can be used.



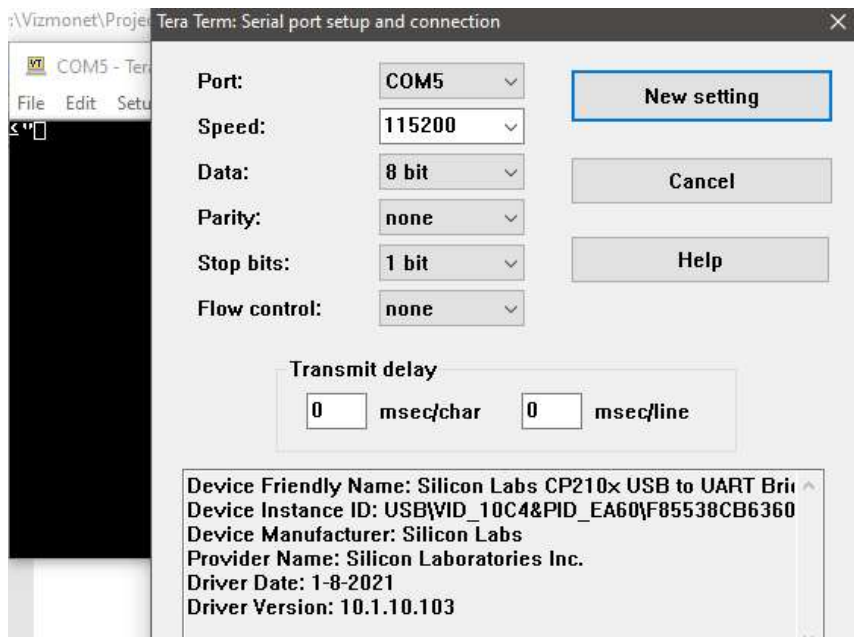
7. Please open the COM/Console cable window.

Here we show the **Tera Term COM putty** as an example.

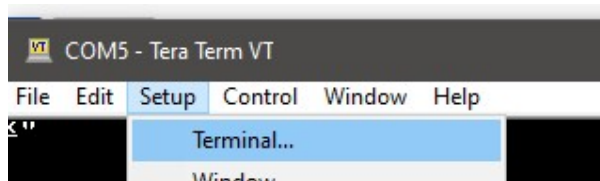
- a. Open the Tera Term COM Putty
- b. Go to 'Setup' and select 'Serial Port'



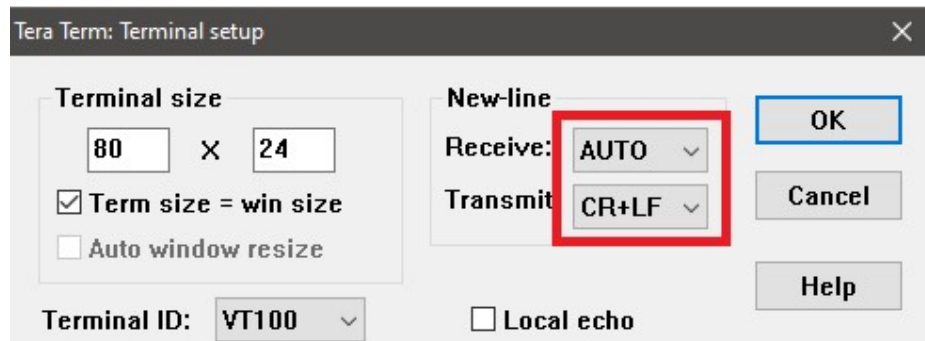
- c. Select the Port and Set the baud rate as 115200



- d. In the setup -> go to Terminal settings



- e. Select the Receive and Transmit as 'Auto' and 'CR+LF' respectively as shown in below figure



8. Now the messages from Spresense board will be displayed here in the console window as shown below

```
NuttShell <NSH> NuttX-10.2.0
nsh>
```

2.3 Wi-Fi HaLow Initial Configurations

1. Please Power ON the board and wait for the **nsh>** prompt in the console window

```
NuttShell (NSH) NuttX-10.2.0
nsh>
```

2. Please type **wiHaLow** and press the **enter** button to start the Wi-Fi HaLow, as shown in below figure

```
nsh> wihalo
Starting WiFi HALO service / utility
[mbpro.local-Fri Jun 30 11:26:54 2023]
[WH1728]
Successfully read configuration
SYS
e2 08 92 72 01 00 00 00 00 04 01 00 02 a0 92 72 : ...r.....r
EIRQ
00 04 : ..
STAT
00 00 00 00 00 00 00 00 20 80 00 00 10 : .....
MSG
4e 52 43 2d 48 53 50 49 00 02 20 00 00 02 20 00 : NRC-HSPI.. ...
[ HSPI SYS Registers ]
- Status      : Ready(0), Sleep(0)
- Chip ID     : 7292
- Modem ID    : 00000001
- Software ID : 00010400
- Board ID    : 7292A002
[ HSPI EIRQ Registers ]
- mode       : active(low), trigger(level), disable
- enable     : txq(0), rxq(0), ready(1), sleep(0)
[ HSPI STATUS Registers ]
- latch      : 0x00
- eirq       : txque(0), rxque(0), ready(0), sleep(0)
- txq        : error(0x00), slot_cnt(0), slot_size(0/0)
- rxq        : error(0x00), slot_cnt(32), slot_size(128/4096)
WiHalo service initialized
```

3. Press the **tab** button in the keyboard, then available commands will be displayed as shown below

```
Options: a / s / r / C / A / S / E / e / Q / q / h
```

Enter the key **h** to get the information about those commands

```
Supported comands
a      - Start access point
s      - Start station
C      - Enter a wihalo lib command or ATCMD
S      - Start a TCP stream server, port 128
A      - Start a TCP stream client, port 128
E      - Start a UDP echo server, port 129
e      - Start a UDP echo client, port 129
Q      - Terminate camera client / server application
q      - Quit the wihalo application
```

4. To see the existing configuration please follow below steps

- a. Press **tab** button in the keyboard

```
Options: a / s / r / C / A / S / E / e / Q / q / h
```

- b. Type **C**

```
Options: a / s / r / C / A / S / E / e / Q / q / h
```

```
Enter command : █
```

- c. Type **cfg show** and then press enter. It will display the existing configuration

```
Current Configuration
SPI clock : 20000000
Country   : JP
DHCP      : ON
Security  : NONE
SSID      : wihalo-vizmo
IP Params : 192.168.1.1,255.255.255.0,192.168.1.1
Flags     : 40010
Channel/bw : 924.500000004
AT delay  : 0
```

5. To change/setup the SSID,

- a. Please press the **tab** button and
- b. then Type **C**
- c. Type **cfg ssid desired_ssid_name** and then press enter

```
Options: a / s / r / C / A / S / E / e / Q / q / h
```

```
Enter command : cfg ssid wihalo-vizmo
```

```
Set SSID to wihalo-vizmo
```

If it was successfully set the SSID, then **Set SSID to** message will be displayed

6. To change/setup the Country,

- a. Please press the **tab** button and
- b. then Type **C**
- c. Type **cfg country JP** and then press enter

```
Options: a / s / r / C / A / S / E / e / Q / q / h
```

```
Enter command : cfg country JP
```

```
Set country code to JP
```

Note: Both Access Point and Station should have same SSID name, Country Code to get connected each other. So please do the steps 5 and 6 on both access point and station boards

7. To change/setup the Frequency,
 - a. Please press the **tab** button and
 - b. then Type **C**
 - c. Type **cfg frequency 924.5** and then press enter

```
Options: a / s / r / C / A / S / E / e / Q / q / h
Enter command : cfg frequency 924.5
Set frequency to 924.500000 Mhz
```

8. To change/setup the Bandwidth,
 - a. Please press the **tab** button and
 - b. then Type **C**
 - c. Type **bw 4** and then press enter

```
Options: a / s / r / C / A / S / E / e / Q / q / h
Enter command : bw 4
Bandwidth spec is now 4 Mhz
```

Note: Please ensure the set BW belongs to the frequency selected. Please refer the Appendix section 3.1 for JP supported frequencies and their corresponding bandwidths.

9. Once the configuration is changed (either SSID/Country Code/Frequency/BW), then the changes should be saved.
 - a. Please press the **tab** button and
 - b. then Type **C**
 - c. Type **save** and then press enter

```
Options: a / s / r / C / A / S / E / e / Q / q / h
Enter command : save
saved default configuration
```

10. Then please **power OFF and then ON the board**, for the new saved configurations to take effect

2.4 Access Point and Station setup

1. We need two DUTs, one for Access point, let's called as DUT1.
Another DUT for Station, let's called as DUT2
2. Please Power ON the DUT and wait for the **nsh>** prompt in the console window

```
NuttShell <NSH> NuttX-10.2.0
nsh>
```

Please do this step for both DUT1 and DUT2

3. Please type **wiHaLow** and press the **enter** button to start the Wi-Fi HaLow, as shown in below figure

```
nsh> wihalo
Starting WiFi HALO service / utility
[mbpro.local-Fri Jun 30 11:26:54 2023]
[WH1728]
Successfully read configuration
SYS
e2 08 92 72 01 00 00 00 00 04 01 00 02 a0 92 72 : ...r.....r
EIRQ
00 04 : ..
STAT
00 00 00 00 00 00 00 00 00 20 80 00 00 10 : .....
MSG
4e 52 43 2d 48 53 50 49 00 02 20 00 00 02 20 00 : NRC-HSPI.. ...
[ HSPI SYS Registers ]
- Status : Ready<0>, Sleep<0>
- Chip ID : 7292
- Modem ID : 00000001
- Software ID : 00010400
- Board ID : 7292A002
[ HSPI EIRQ Registers ]
- mode : active<low>, trigger<level>, disable
- enable : txq<0>, rxq<0>, ready<1>, sleep<0>
[ HSPI STATUS Registers ]
- latch : 0x00
- eirq : txque<0>, rxque<0>, ready<0>, sleep<0>
- txq : error<0x00>, slot_cnt<0>, slot_size<0/0>
- rxq : error<0x00>, slot_cnt<32>, slot_size<128/4096>
WiHalo service initialized
```

Please do this step for both DUT1 and DUT2

4. Press the **tab** button and then enter **a** to start the access point with DUT1

And it will display below message

```
Options: a / s / r / C / A / S / E / e / Q / q / h

Configuring NRC device
Started the access point
Setting IP parameters
Started DHCP server
```

5. Press the **tab** button and then enter **s** to start the station on DUT2

And it will display below message

```
Options: a / s / r / C / A / S / E / e / Q / q / h
Resetting modem..
Device has been reset
Connecting..
Connected to the access point
Event - DHCP_START processed
Event - DHCP_SUCCESS processed
```

2.5 Server and Client setup

1. Please set one DUT as access point and another DUT as station by following the procedures in Section 2.4

2. To set one DUT as server, Press the **tab** button and then enter **S** on DUT1

```
Options: a / s / r / C / A / S / E / e / Q / q / h
Starting server thread
Waiting for connection
```

3. To set another DUT as client, Press the **tab** button and then enter **A** on DUT2.

It will ask for the IP address, please enter the Server DUT's IP address and then press enter

```
Options: a / s / r / C / A / S / E / e / Q / q / h
Enter Camera IP address: 192.168.1.1
```

(Note: Access Point's IP address is 192.168.1.1 which is DUT1 and also configured as server here and the station's IP address is 192.168.1.2 which is DUT2 configured as client here)

4. Now the communication starts between the server and client and throughput is displayed on the client side as shown in below figure.

```
Options: a / s / r / C / A / S / E / e / Q / q / h
Enter Camera IP address: 192.168.1.1
Total: 36921328 bytes / Data Rate: 3323523 bps
```

It is 3.3Mbps here in the picture.

5. Press **tab** and then enter **Q** to stop the test either on client or server side

```
Options: a / s / r / C / A / S / E / e / Q / q / h
Total: 71310968 bytes / Data Rate: 3342342 bps
Total: 71725136 bytes / Data Rate: 3341978 bps
Closing connection
Done
```

2.6 Latency Test

1. Please set one DUT as access point and another DUT as station by following the procedures in Section 2.4

2. To set one DUT as server, Press the **tab** button and then enter **E** on DUT1

```
Options: a / s / r / C / A / S / E / e / Q / q / h
Waiting for echo_requests
```

3. To set one DUT as client, Press the **tab** button and then enter **e** on DUT2

```
Options: a / s / r / C / A / S / E / e / Q / q / h
Enter Echo server IP address: 
```

4. Please enter the IP address as 192.168.1.1

```
Options: a / s / r / C / A / S / E / e / Q / q / h
Enter Echo server IP address: 192.168.1.1
```

(Note: Access Point's IP address is 192.168.1.1 which is DUT1 and also configured as server here and the station's IP address is 192.168.1.2 which is DUT2 configured as client here)

5. Now the communication starts between the server and client and the latency is displayed on the client side as shown in below figure.

```
Echo response received <258>, Latency: 0.0219 secs
Echo response received <259>, Latency: 0.0171 secs
Echo response received <260>, Latency: 0.0175 secs
Echo response received <261>, Latency: 0.0165 secs
Echo response received <262>, Latency: 0.0177 secs
Echo response received <263>, Latency: 0.0172 secs
Echo response received <264>, Latency: 0.0171 secs
Echo response received <265>, Latency: 0.0167 secs
Echo response received <266>, Latency: 0.0167 secs
Echo response received <267>, Latency: 0.0191 secs
Echo response received <268>, Latency: 0.0176 secs
Echo response received <269>, Latency: 0.0214 secs
```

6. Press **tab** and then enter **Q** to stop the test either on client or server side

```
Echo response received <939>, Latency: 0.0186 secs
Closing echo client thread
Done
```

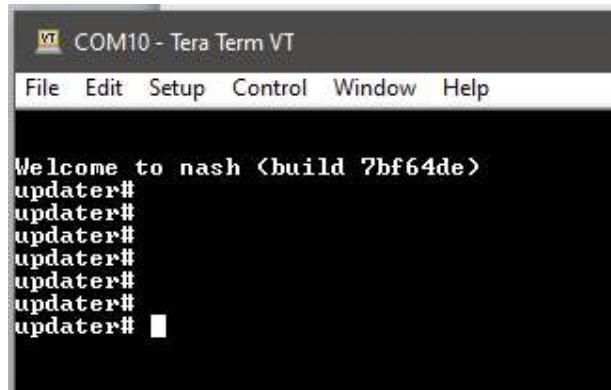
3. APPENDIX

3.1 JP Bands for Wi-Fi HaLow

BW(MHz)	Frequency(MHz)
1	921
1	923
1	924
1	925
1	926
1	927
2	923.5
2	924.5
2	925.5
2	926.5
4	924.5
4	925.5

3.2 Program the image on Sony Spresense

1. Once power ON the board, below will be displayed in the COM port of the SONY for the brand new Sony Main board.



```
VT COM10 - Tera Term VT
File Edit Setup Control Window Help

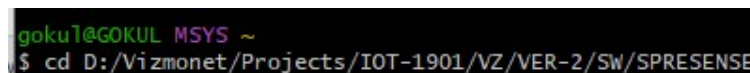
Welcome to nash (build 7bf64de)
updater#
updater#
updater#
updater#
updater#
updater#
updater#
```

To make it work, we have to program the bootloader and nuttx.spk.

2. If PC does not have MSYS2 tool , then please download the tool 'msys2-x86_64-20220603.exe' from the MSYS2 website and install the MSYS software. This software is used to program the flash in nuttx/linux environment from the windows PC.
3. Copy the loader.espk and nuttx.spk into the 'SPRESENSE' folder of the Spresense-mainboard. Attached the zip files with required files in place, but please get the latest nuttx.spk file. Please download and extract below attachment.



4. Open the MSYS window and go to that 'SPRESENSE' directory as shown on below figure.



```
gokul@GOKUL MSYS ~
$ cd D:/Vizmonet/Projects/IOT-1901/VZ/VER-2/SW/SPRESENSE
```

5. Execute below command to flash the boot loader (If already bootloader installed, please skip this step)

tools/flash.sh -c COM6 loader.espk

```
gokul@GOKUL MSYS /d/Vizmonet/Projects/IOT-1901/VZ/VER-2/SW/SPRESENSE
$ tools/flash.sh -c COM6 loader.espk
/usr/bin/env: 'python3': No such file or directory
>>> Install files ...
install -b 115200
Install loader.espk
|0%-----50%-----100%|
#####
129968 bytes loaded.
Package validation is OK.
Saving package to "loader"
updater# sync
updater# Restarting the board ...
reboot
```

6. Execute below command to flash the nuttx ATCMD application.

tools/flash.sh -c COM6 nuttx.spk

```
gokul@GOKUL MSYS /d/Vizmonet/Projects/IOT-1901/VZ/VER-2/SW/SPRESENSE
$ tools/flash.sh -c COM6 nuttx.spk
/usr/bin/env: 'python3': No such file or directory
>>> Install files ...
install -b 115200
Install nuttx.spk
|0%-----50%-----100%|
#####
205648 bytes loaded.
Package validation is OK.
Saving package to "nuttx"
updater# sync
updater# Restarting the board ...
reboot
```

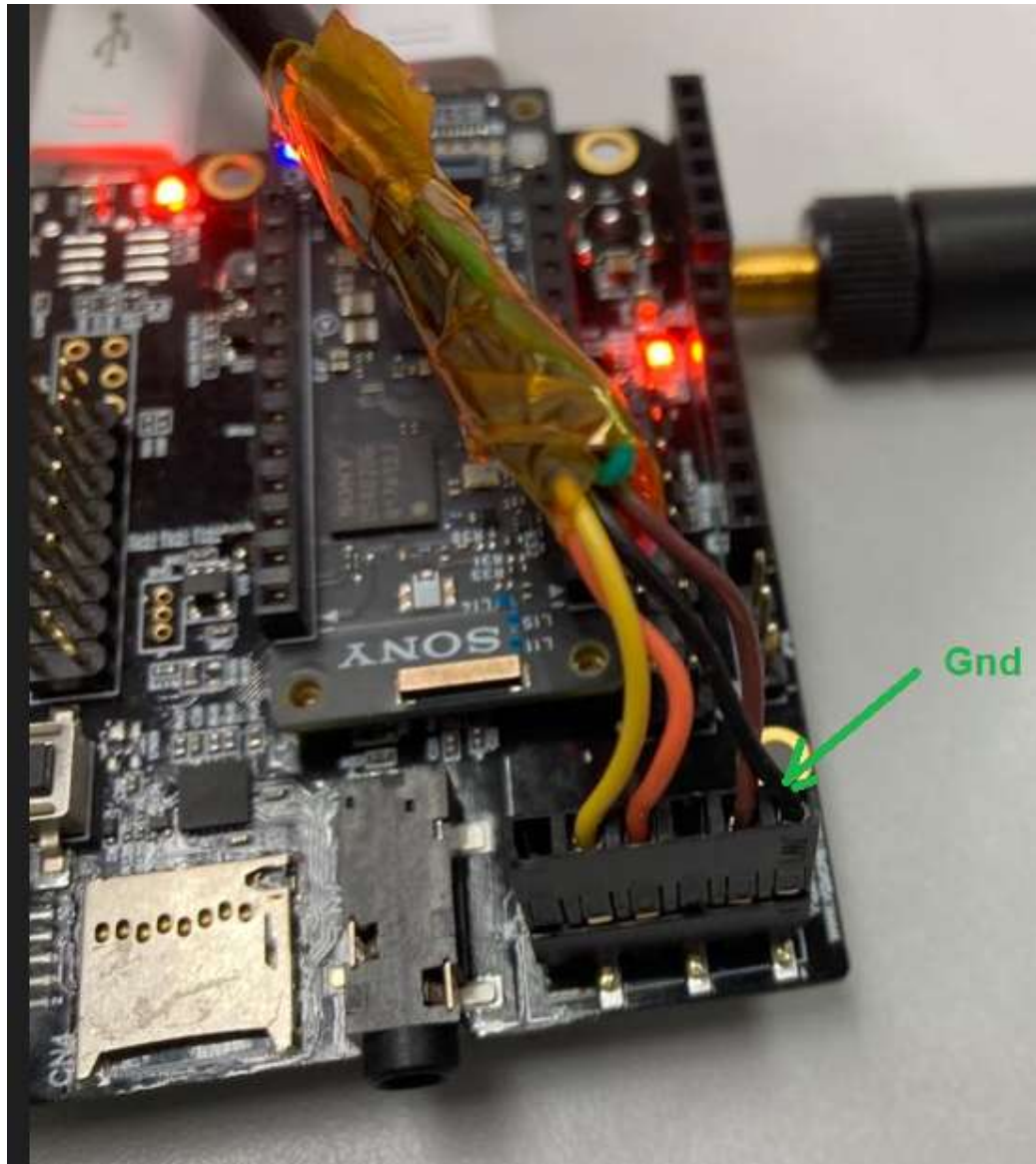
3.3 Wi-Fi HaLow Card Console

1. There is one console cable header (6-pins header out of which 4 pins are used) for Wi-Fi HaLow card for debug purpose. It can be used if required.

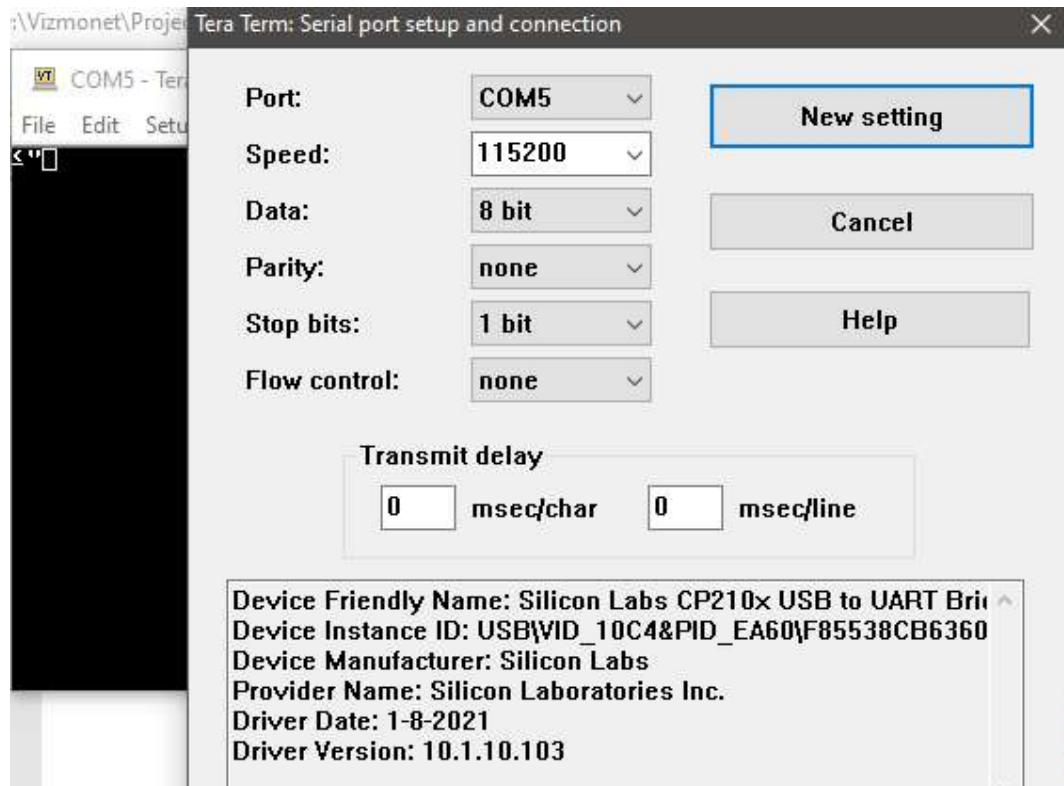
(Please remove the Red (power) and green or blue wires from the default USB console cable before connecting with this header)

The ground polarity and the cable orientation is as shown in below figure.

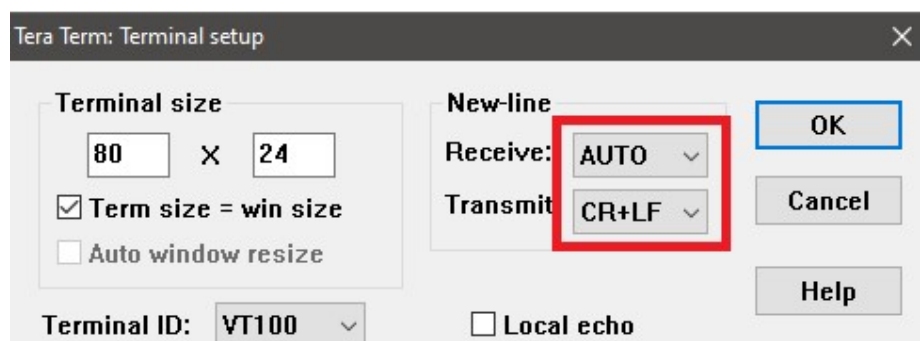
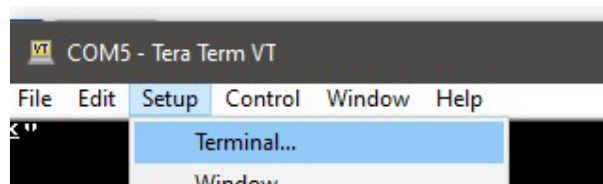
Pins 3 and 6 are no connection from right as in below figure.



2. Please set the baud rate as 115200



3. In the setup → Terminal settings, select the Receive and Transmit as 'Auto' and 'CR+LF' respectively as shown in below figure



4. Now, the messages from Wi-Fi HaLow card will be displayed here in the window as shown below.

```

VT COM3 - Tera Term VT
File Edit Setup Control Window Help
Inrc_i2c_configl using peripheral clock(80000000). setting clock=4000000, with=0
Inrc_i2c_configl calculated prescale=3
temp_sensor: Temperature sensor is a TMP310B [Address : 0x71]
Boot Reason: 000000 [POR] 000000
Cold Boot. Skip Recovey
wlif_init
net: netif_is_up, local interface IP is 0.0.0.0

wlif_init
net: netif_is_up, local interface IP is 0.0.0.0

[ATCMD] ATCMD_BUILD: CM0,XIP,GLOBAL,LWIP
[ATCMD] ATCMD_VERSION: 1.22.1
[ATCMD] ATCMD_TASK_PRIORITY: 2
[ATCMD] ATCMD_MSG_LEN_MIN: 2
[ATCMD] ATCMD_MSG_LEN_MAX: 128
[ATCMD] ATCMD_DATA_LEN_MAX: 4096
[ATCMD] ATCMD_TXBUF_SIZE: 4096
[ATCMD] ATCMD_RXBUF_SIZE: 4096
[ATCMD] ATCMD_CFG_PROMPT_OFF
[ATCMD] ATCMD_CFG_ECHO_OFF
[ATCMD] ATCMD_CFG_HISTORY_OFF
[ATCMD] ATCMD_CFG_LOWERCASE_OFF
[ATCMD] ATCMD_CFG_LINEFEED_ON
[ATCMD] ATCMD_SOCKET_DIRECT_SEND
[ATCMD] RF_CAL_INFO: cal_use=1 country=US
[ctrl_iface_receive_response] cmd: set country US
reply_len: 3
reply: OK

-- Country code: 'US'-- Country c, number of code: 'US', n channel:45umber of cha
nel:45
[ATHIF] HSPI_OPEN: sw_id=0x00010304 bd_id=0x7292A002
[ATHIF] HSPI_FIFO: rx=(0x1043a410, 16384), tx=(0x1043e4a0, 16384)
[ATHIF] HSPI_MSG_REG:
[ATHIF] - msg[0]: 0x2D43524E, NRC- <ID0>
[ATHIF] - msg[1]: 0x49505348, HSPI <ID1>
[ATHIF] - msg[2]: 0x00200200, slot_num=32, slot_size=512 <TXQ>
[ATHIF] - msg[3]: 0x00200200, slot_num=32 slot_size=512 <RXQ>
[ATHIF]

[0] 6757483 nrc7292_standalone_xip>
[1] 7698839 nrc7292_standalone_xip>

```